

EnergyAI layered architecture

Experience layer

Next.js 16 / React 19
Dashboard, Usage, Forecast, Actions, Settings

API and policy layer

FastAPI routers, JWT sessions, RBAC, validation

Domain services

Forecasting, ingestion, analytics, alerts, email, account lifecycle

Persistence and background work

PostgreSQL 16; simulation, alert,
email, and avatar-cleanup workers

Model artifacts

Global TFT 24 h and 168 h, manifests, hashes, readiness gates